

Lattice QCD Notes

Shivam Panda

1 Background

Typically, the standard Hamiltonian is the generator of evolution. So, in typical QM, the transition amplitudes (the probability of a particle starting at x_i at t_i to be found at x_f at t_f is given by:

$$\langle x_f | e^{-iH(t_f - t_i)} | x_i \rangle \quad (1)$$

However, the imaginary exponential leads to computational troubles due to rapid oscillations, so a Wick Rotation is performed with transforms $it \rightarrow t$. This Wick Rotation switches Minkowski space to Euclidean space, where time behaves as another spacial dimension and allows the computation using a 4D grid. However, this change in time causes significant limitations on simulating dynamics and time-evolution in the lattice. This leads to exponential decrease rather than oscillations, which is much more computationally stable. With this transformation, the transition amplitudes can be written as:

$$\langle x_f | e^{-H(t_f - t_i)} | x_i \rangle = \int Dx(t) e^{-S[x]} \quad (2)$$

This is a 1-dimensional path integral, where $Dx(t) = A \int dx_1 dx_2 dx_3 \dots dx_{N-1}$ as $N \rightarrow \infty$. Each dx_i represents all the possible positions at different slices of time. Essentially, it is an integral of all possible functions of $x(t)$. Paths with higher action are exponentially suppressed, which means that the leading contributor is the path with minimal action. After the Wick Rotation, the action and Lagrangian change

$$S[x] = \int dt L(x, \dot{x}) = \int dt \left[\frac{m\dot{x}^2}{2} + V(x(t)) \right] \quad (3)$$

The most significant difference is the positive sign in the Lagrangian. Using this formulation, the propagator can be written by resolving the identity into energy eigenstates. Since the Hamiltonian is diagonal with the energy eigenstates, the exponential simplifies from the Hamiltonian to simple energy eigenvalues.

$$\langle x | e^{-HT} | x \rangle = \sum \langle x | E_i \rangle e^{-E_i T} \langle E_i | x \rangle \quad (4)$$

Due to the exponential suppression, as $T \rightarrow \infty$, only the ground state remains and $\langle x | e^{-HT} | x \rangle = e^{-E_0 T} |\langle x | E_0 \rangle|^2$. Since $\psi_n(x) = \langle x | E_n \rangle$, which is just an energy eigenstate written in the position basis. The normalization properties of the wavefunction lead to $\int |\psi_n(x)|^2 dx = 1$, so that the ground state can be extracted by integrating all x .

$$\int dx \langle x | e^{-HT} | x \rangle = e^{-E_0 T} \quad (5)$$

However, this equation only holds true as $T \rightarrow \infty$

1.1 Discrete Path Integral

Since the computer cannot simulate all possible functions $x(t)$, it is possible to make the t -axis discrete, and describe the position $x(t_i)$, and integrate with all possible x to replicate the idea of integrating over all possible functions. Therefore, there are as many integrals as there are grid points on the t -axis. The t -axis can be made discrete by $t_j = t_0 + ja$, where $a = \frac{t_f - t_0}{N}$. With this, a path described as a vector $x = \{x(t_0), x(t_1), \dots, x(t_N)\}$. Since the end points are fixed, the path integral becomes

$Dx(t) = A \int dx_1 dx_2 dx_3 \dots dx_{N-1}$ as $N \rightarrow \infty$. A is simply a normalization factor. Using this, the action integral can be determined using discrete approximations of the derivative.

$$\int_{t_j}^{t_{j+1}} dt = \int_{t_j}^{t_{j+1}} a \left[\frac{m}{2} \left(\frac{x_{j+1} - x_j}{a} \right)^2 + \frac{V(x_{j+1}) + V(x_j)}{2} \right] \quad (6)$$

The derivative is replaced by the limiting definition of the derivative, where dt is taken as a , and the potential is determined at an interval by taking the average value. Using all of this, the propagator and path integral can be computed.

$$\langle x | e^{-HT} | x \rangle = A \int_{-\infty}^{\infty} dx_1 dx_2 \dots dx_{N-1} e^{-S_{lat}[x]} \quad (7)$$

$$S_{lat}[x] = \sum_{j=0}^{N-1} \left[\frac{m}{2a} (x_{j+1} - x_j)^2 + aV(x_j) \right] \quad (8)$$

1.2 Monte Carlo Evaluation

In the previous section, the probability amplitude was a path integral, however the every path had the same boundary conditions. However, in this next section, the path integrals will be evaluated at every possible boundary condition, so it is effectively an integral of path integrals in some sense. To find the correlation between $x(t_2)$ and $x(t_1)$,

$$\langle\langle x(t_2)x(t_1) \rangle\rangle = \frac{\int Dx(t)x(t_2)x(t_1)e^{-S[x]}}{\int Dx(t)e^{-S[x]}} \quad (9)$$

Here, both path integrals are integrating over all paths for a set of boundary conditions, and then also integrating over all possible boundary conditions. In general, the $\langle\langle x(t_1) \rangle\rangle$ gives information about on average, what is the position of the particle at t_1 , where the average is taken over all paths. The correlation, like shown in Eq (9), gives information about the correlation between the position of the particle at t_1 and t_2 . If $t_1 = t_2$, then this would give information about the variance of $x(t_1)$, which is essentially the width of the position due to quantum fluctuations. Since the integral is over all possible boundary conditions, the denominator can be written as

$$\int Dx(t)e^{-S[x]} = \int dx \langle x | e^{-HT} | x \rangle = \sum e^{-E_n T} \quad (10)$$

This integral over boundary condition is the trace of an operator, which is just the sum of its eigenvalues. Going from this integral to sum is simply changing the basis for this trace calculation, moving from position to energy basis. This equation is considered the partition function for a system. The numerator can be simplified by considering that $t_f - t_1 = T$ and $t = t_2 - t_1$. A similar approach can be taken with the numerator of Eq 9, but more care needs to be taken.

$$\sum_n |E_n\rangle e^{-H(t_f - t_2)} \hat{x} e^{-H(t_2 - t_1)} \hat{x} e^{-H(t_1 - t)} \langle E_n| \quad (11)$$

This expression rewrites the operator in energy basis, but it also gives intuition as to what the operator actually does. Essentially, it allows the system to evolve from t_i to t_1 , then applies the position operator. Then, allows the system to evolve from t_1 to t_2 , and then applies another position operator at t_2 . Finally, it allows the system to evolve for the rest of the time considered. By switching to the energy basis, the Hamiltonian becomes diagonal, so the evolution terms simplify.

$$\sum_n e^{-E_n(t_f - t_2 + t_1 - t_i)} \langle E_n | \hat{x} e^{-H(t_2 - t_1)} \hat{x} | E_n \rangle \quad (12)$$

$$\sum_n e^{-E_n T} \langle E_n | \hat{x} e^{-(H - E_n)t} \hat{x} | E_n \rangle \quad (13)$$

Using this equation, and assuming that $T \gg t$, where T is very large, the correlation

$$G(t) = \frac{e^{-E_0 T} \langle E_0 | \hat{x} e^{-(H - E_0)t} \hat{x} | E_0 \rangle}{e^{-E_0 T}} = \langle E_0 | \hat{x} e^{-(H - E_0)t} \hat{x} | E_0 \rangle \quad (14)$$

Since, as T becomes very large, the ground state dominates. Using parity arguments, the position operator is odd, while the $|E_0\rangle$ is even. Therefore, as t is large,

$$G(t) = |\langle E_0|\hat{x}|E_1\rangle|^2 e^{-(E_1-E_0)t} \quad (15)$$

Using this formulation, the first energy eigenstate can be determined by evaluating the correlation function at t and $t+a$.

$$\log\left(\frac{G(t)}{G(t+a)}\right) = (E_1 - E_0)a \quad (16)$$

$\langle E_0|\hat{x}|E_1\rangle$ the transition matrix element that physically determines the probability of the position operator kicking the first energy eigenstate into the ground state. Using this technique, $\langle\langle \Gamma[x] \rangle\rangle$ can be determined by generating a large number of configurations (paths) on the grid such that the probability of obtaining a particular path is proportional to $e^{-S[x]}$, and thus, the expected value of $\Gamma[x]$ is simply a weighted average of these configurations. This is a Monte Carlo estimator for the true expected value of the functional, however the uncertainty decreases with $1/\sqrt{N}$.

Generating paths with this probability proportional to the exponent of the action is not a simply task however it can be performed by the Metropolis Algorithm.

$$\langle\langle \Gamma[x] \rangle\rangle = \frac{1}{N_{cf}} \sum_{\alpha}^{N_{cf}} \Gamma[x^{(\alpha)}] \quad (17)$$

This begins by start with a random path x^0 . Then to generate x^1 , all of the sites of the lattice are visited and the values at those sites are randomized by this procedure.

- 1) Generate a random number $(\alpha) \in [-\epsilon, \epsilon]$
- 2) Replace x_j with $x_j + \alpha$
- 3) If action is reduced, actually replace x_j with this new value
- 4) If action is increased, then generate a random number (β) between 0 and 1. Retain new value if $e^{-\Delta S} > \beta$. Otherwise, keep old value and move to next site.

The value of ϵ is critical since if it is too high, then most new values for x_j will be rejected, while if it is too small, many will get accepted, but the values will not change significantly. The goal is that x_j should pass with probability 0.4 - 0.6. Additionally, paths should be systematically chosen as to eliminate the correlation between successive paths. Lastly, the first $5N$ or $10N$ should be disregarded since it is extremely random and useless.

1.3 Statistical Errors

It is incredibly computationally intensive to estimate the statistical uncertainty of these monte carlo estimators by repeating the entire estimate a 100 times. Therefore, there is Statistical Bootstrapping. Essentially, from the given values of $G(t)$ for each path, a random selection of these $G(t)$'s can be picked to find an estimate, and these estimates can be used to determine the statistical errors. Another useful process is called binning, where a systematic sampling of configurations and their corresponding $G(t)$ values can be taken to find estimates and therefore be partially averaged to get estimates. These estimates can then be used to determine the statistical errors. One key limitation of binning is that within a bin, the configurations could possibly be correlated, and therefore, the errors estimated could be much lower than the true errors. This would occur if N_{corr} is taken to be too small.

2 Field Theory on Lattice

With transition from QM to Field Theory is essentially adding spatial dimension. The $x(t)$ from before could be considered a field theory of 0 spatial and 1 temporal dimension. Instead of have a ground state energy eigenstate, the field has a ground state $|0\rangle$, which corresopnds to no particles. The operators ϕ behaves like the position operator, and it is possible for this operator to excite the field above its ground state, and create a particle.

The nodes of the 4D Euclidean grid, separated by the lattice spacing a , all have corresponding values of the field. The partition function still exists in path integral form:

$$Z = \int e^{-S[\phi]} \prod d\phi(x_j) \quad (18)$$

This is essentially a path integral over all possible field configurations. $S[\phi]$ corresponds to the field action. The spatial and temporal derivatives transform in the differences of the scalar field with neighboring field values. Similar to the operators used in QM to study excitations from the ground state, operators like:

$$\Gamma(t) = \frac{1}{\sqrt{N}} \sum \phi(\vec{x}_j, t) \quad (19)$$

Summing over all the spatial vectors enforces zero 3-momentum. Since the excitations correspond to particle creation, their energies are the masses of the particles. Any non-zero momentum would destructively interfere and cancel itself out since it would repeat throughout the lattice. (QUESTION)

$$\langle\langle \Gamma(t)\Gamma(0) \rangle\rangle = | \langle 0 | \Gamma(0) | \phi : \vec{p} = 0 \rangle |^2 e^{-m_\phi t} \quad (20)$$

Where $|\phi : \vec{p} = 0 \rangle$ represents a single ϕ -particle state with zero three momentum and m_ϕ represents the mass of the particle.

Often, additional terms are added into the Lagrangian to add local interactions to account for the ultraviolet cutoff. The ultraviolet cutoff is simply the lattice property that there is a minimum wavelength ($\lambda = 2a$), where a is the lattice spacing. Sometimes, these additional terms influence new modes of oscillations and solutions which are considered "numerical ghosts".

To improve these discretization errors, it is possible to make the transformation

$$x_j = \tilde{x}_j + \psi_1 a^2 \Delta^2 \tilde{x}_j + \psi_2 a^2 \omega_0^2 \tilde{x}_j \quad (21)$$

With the correct choices of ψ_1 and ψ_2 , all $O(a^2)$ terms can cancel in the action, and the discretization errors will be $O(a^4)$ and all numerical ghosts can be removed.

3 QCD on a Lattice